

AA6 - Conversion from RE to FA

(soluções)

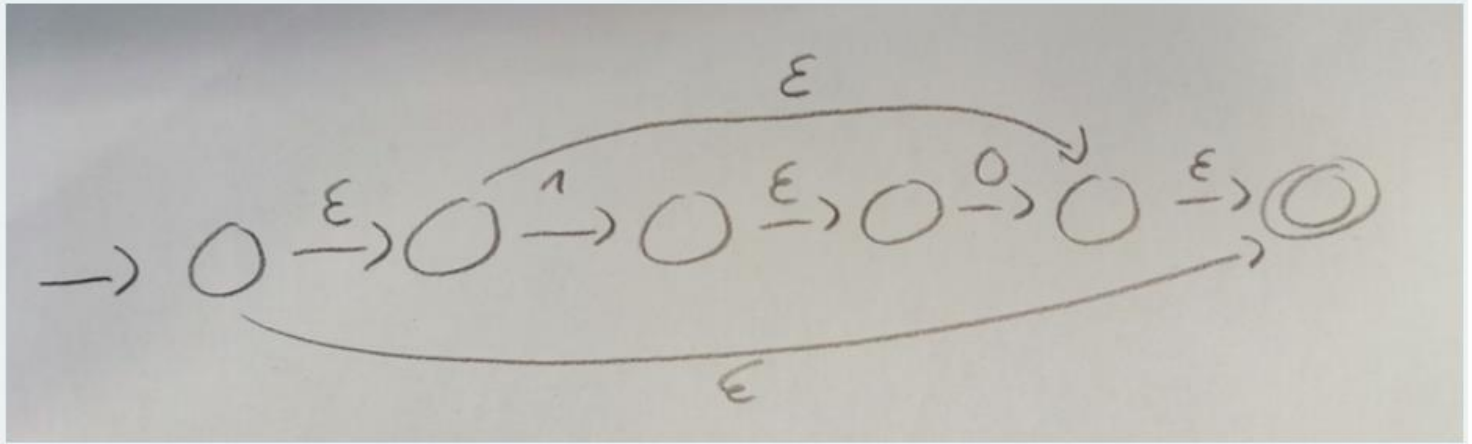
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Given the regular expressions re1, re2 and re3, with corresponding automatas fa1, fa2 and fa3, which of the following options describes the correct way to build an automata (following the rules to draw finite automatas) that can recognize the union of the languages defined by re1, re2 and re3?

(1/1 Ponto)

- ☐ First we build the automata containing fa1 and fa2, representing the union of re1 with re2. Then we add fa3 to the automata, representing the union of re1 + re2 with re3.
- ☐ First we build the automata containing fa2 and fa3, representing the union of re2 with re3. Then we add fa1 to the automata, representing the union of re1 with re2 + re3.
- ☐ First we build the automata containing fa1 and fa3, representing the union of re1 with re3. Then we add fa2 to the automata, representing the union of re1 + re3 with re2.
- ☒ All of the above.

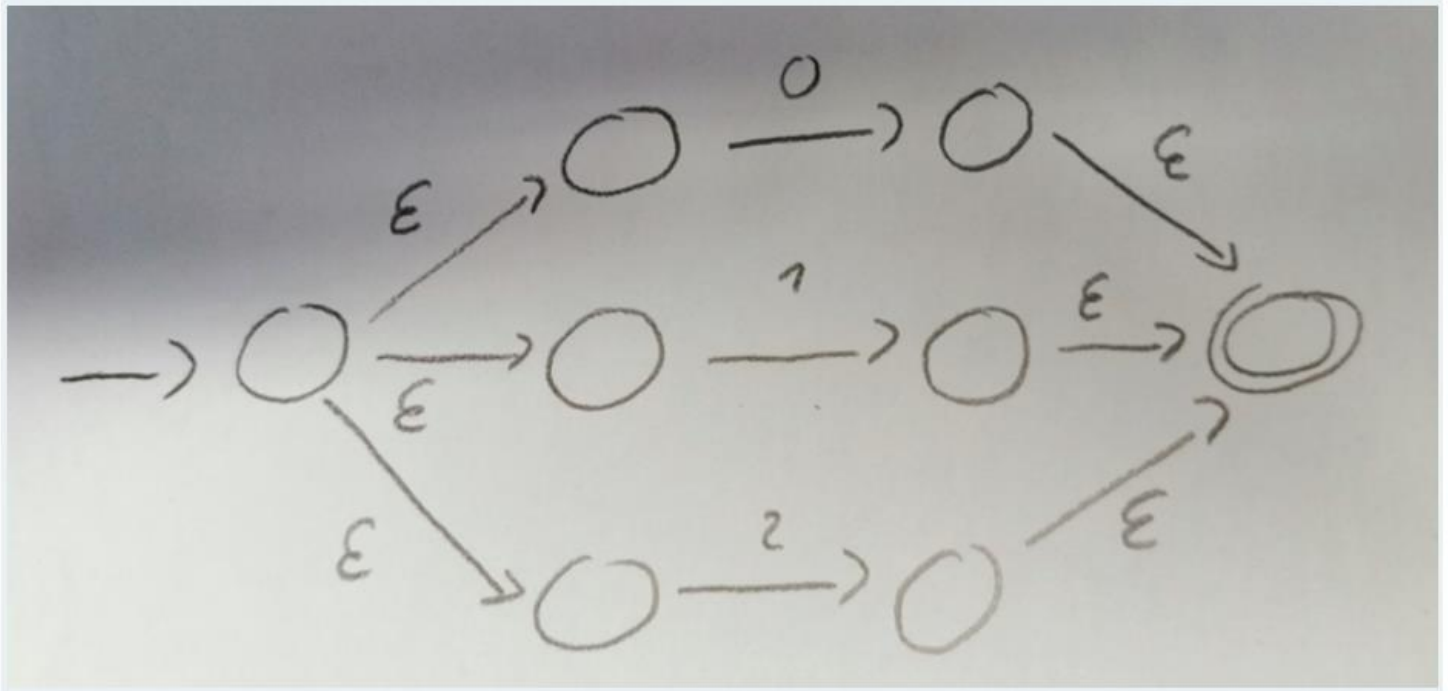
Which of the regular expressions below would originate the following ϵ -NFA (not simplified) by the rules to build a finite automata from a regular expression? 
(1/1 Ponto)



- ☐ 10^*
- ☐ $(10)^*$
- ☐ $\epsilon + (10)^*$
- ☒ None of the above.

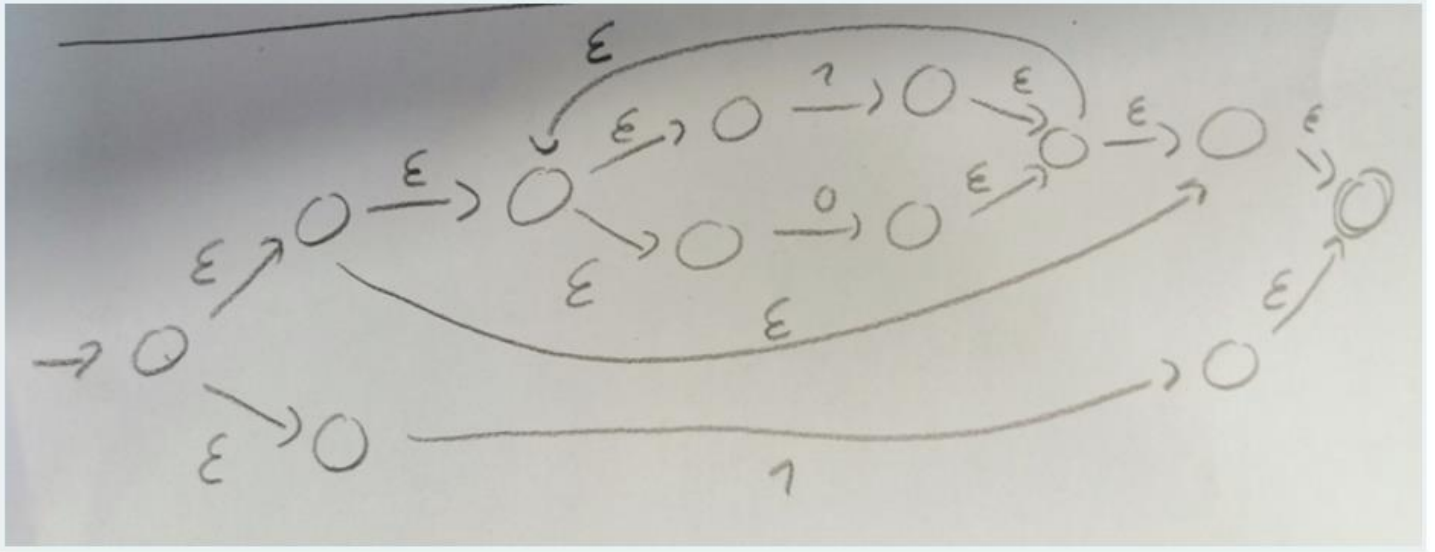
If we strictly apply the rules to build a finite automata from a regular expression, and we do not simplify the result, does the following ϵ -NFA correspond to a correct application of those rules?

(1/1 Ponto)



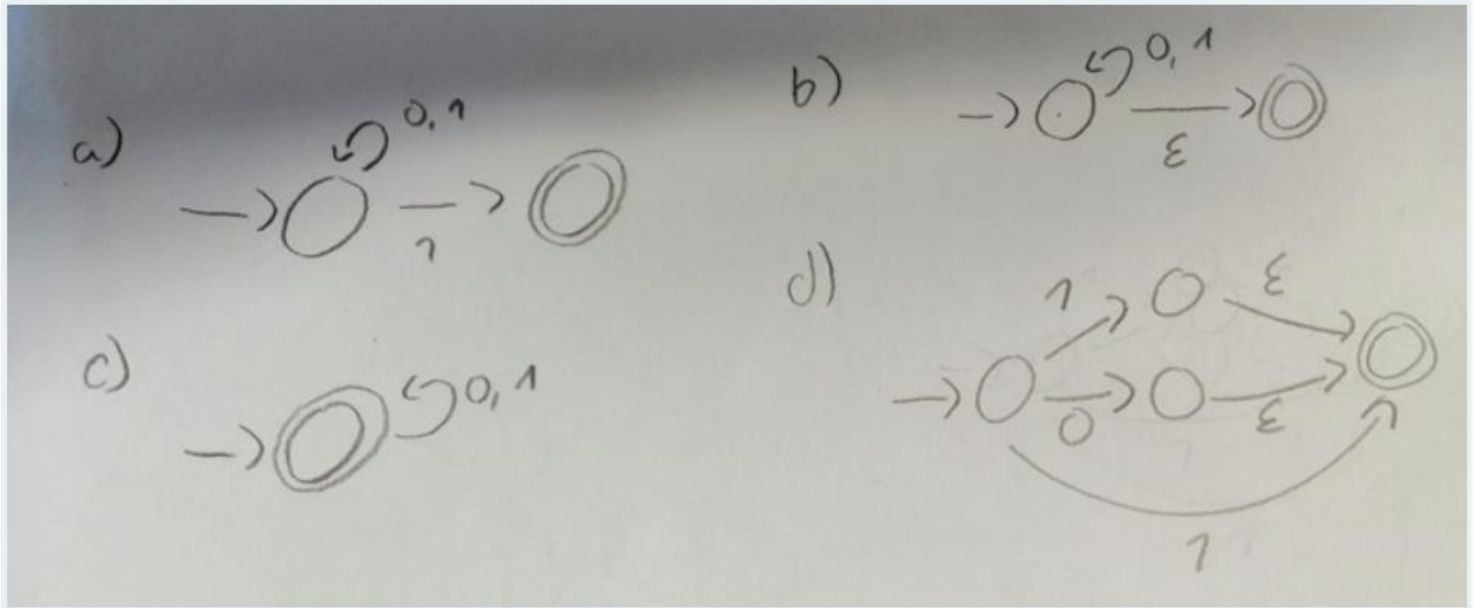
- ☐ Yes, since it corresponds to the union of three finite automata.
- ☒ No, since we would need to first build the union for two finite automatas, and only then add the third.

Which of the following regular expressions generates the following ϵ -NFA (not simplified) when the rules to build a finite automata from a regular expression are applied? 
(1/1 Ponto)



- ☒ $(1 + 0)^* + 1$
- ☐ $(1 + 0)^*$
- ☐ $((1 + 0)^* + \epsilon) + 1$
- ☐ None of the above

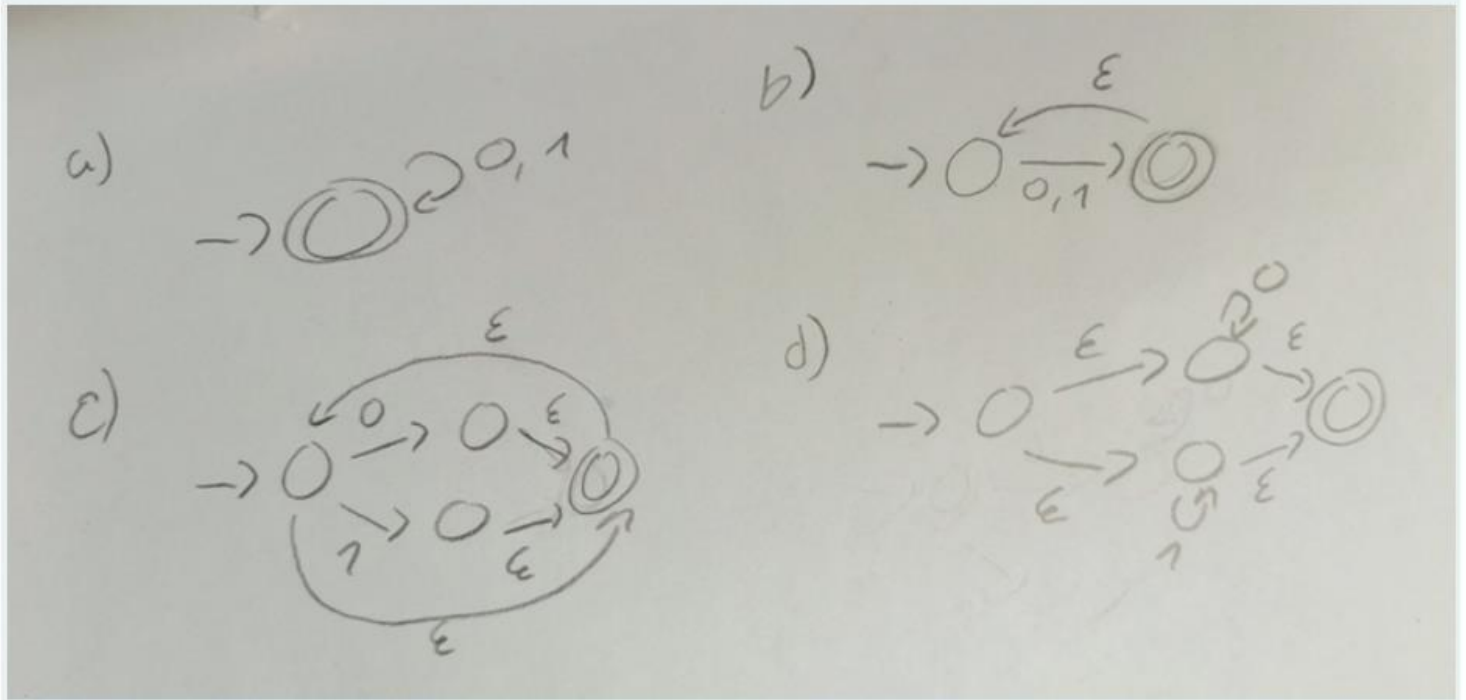
Considering the ϵ -NFA from the previous question, which of the following finite automata would be the result of fully simplifying it?
(2/2 Pontos)



- ☐ FA from option a)
- ☐ FA from option b)
- ☒ FA from option c)
- ☐ FA from option d)
- ☐ None of the above

Which of the following finite automata are equivalent (recognize the same language) among themselves?

(1/1 Ponto)



- ☐ FAs from option a) and b)
- ☐ FAs from option c) and d)
- ☒ FAs from option a) and c)
- ☐ FAs from option b) and d)
- ☐ None are equivalent